



Cars4Mars

African Rover Challenge

Competition full name: **Cars4Mars African Rover Challenge**

Competition short name: **Cars4Mars**

Website www.cars4mars.co.za

Contact person: Ms Basia Nasiorowska – Founder and Organiser basia@cars4mars.co.za

REGISTRATION to participate on website www.cars4mars.co.za

Please note that the Teams have to include the competition name on ALL documentation and ALL Social Media or other advertising material when announcing their participation and winnings. The above competition is the only competition of this kind in Africa and MENA, and has to be acknowledged.

RULEBOOK 2026 – Overview

Note: There may be some updates during the competition, please contact Basia Nasiorowska on basia@cars4mars.co.za

1. Who can compete?

This challenge is addressed to students. You need to compete as a team. All members of the team have to be students actively enrolled in high-school, college or university program during the entire duration of the competition.

Teams are encouraged to consult professors, experts or engineers (called Faculty Advisors) for advice and guidance. However, they can not be registered as part of the team and the engineering, design, manufacturing and operation of the rovers needs to be done by students.

Each team is required to have a single Team Leader (also needs to be a student). This person will be responsible for official contact with the Organizers and making sure that their team complies with the competition rules.

2. What is the challenge?

The challenge is to design and build a Mars rover **prototype**. This is **not expected to be a space-grade machine** and does **not need to be ready to operate in actual space or Mars conditions**.

Instead, the rover prototype should be a small, mobile robot. It needs to be controlled wirelessly and have an independent power source (battery).

The challenge is divided into two stages: Launch Stage (fully online) and Mars Stage (in person in Johannesburg). Each stage is treated as a separate competition, with its own scoring and winners. Only the best teams from the Launch Stage will be accepted to the Mars Stage. However, the points from the Launch Stage are not transferred to the Mars Stage.

If you're not sure if you can travel to Johannesburg for the Mars Stage, don't worry - you can just compete in the Launch Stage and try to win that! However, the Launch Stage is focused on showing that your rover could do well in the missions planned for the Mars Stage - so make sure to check those out and build your rover accordingly, even if you're not planning to come to Johannesburg.

Launch Stage

Launch Stage (1st stage) is held online. During this stage, students will design and build their robots at their homes, schools or universities. They will need to present their project strategy, design and, finally, the capabilities of their manufactured rover.

All of this will be done by sending the judges written reports and a video of the rover.

This stage will have its own final, where teams will have the occasion to meet online, hear interesting talks from guests from the space industry and give their final presentations. The best team of this stage and a full ranking will be officially published.

Mars Stage

Best teams from the Launch Stage (1st stage) will be invited to compete in the Mars Stage (2nd stage). This stage will allow the teams to meet face to face together with their rovers on a specially prepared obstacle course.

Students will need to operate their rovers wirelessly and complete a series of tasks on the special obstacle course. These tasks will be designed to resemble some real scenarios that could be expected from future rovers on the surface of Mars (e.g. exploring rocky terrain, looking for tools left by an astronaut etc.). Top teams from this stage will be awarded.

This special event will provide students an opportunity to not only test their machines in real, thrilling scenarios, but also meet other people with similar passions and invited guests.

3. Competition schedule

Stage	Event	Deadline
Launch Stage (teams design and build rovers in their own towns)	Team registration	30 April 2026
	Project Strategy Report	31 May 2026
	Video + Final Report	2 August 2026
	Launch Stage Final - ONLINE	22 August 2026
Mars Stage (teams drive their rovers together on one course)	Mars Stage Final- IN PERSON Johannesburg South Africa	19 September 2026

4. Scoring

Deliverable / Mission	Max points
Launch Stage	400 pts (Total)
Project Strategy Report	50 pts
Final Design Report	50 pts
Video	200 pts
Launch Stage Final (presentation)	100 pts
Mars Stage	400 pts (Total)
Traversal Mission	250 pts
Autonomous Mission	150 pts

5. Launch Stage

During this stage your team will develop a series of deliverables, that will be sent to judges online. Each of these deliverables is scored and counts towards the final ranking for the Launch Stage.

Timeline of the Launch Stage with deadlines for sending deliverables is presented below:

Event	Deadline
Team registration	30 April 2026
Project Strategy Report	31 May 2026
Video + Final Report	2 August 2026
Launch Stage Presentation	22 August 2026

Each deliverable will be sent online via a form that will be shared with registered teams. The only exception is the final deliverable - Launch Stage Presentation - which will be held live online during a special event, followed by a ceremony to award top teams.

These deliverables are described below.

5.1. Project Strategy Report

This is a written document presenting your team's strategy for a successful run in the challenge and your initial ideas for the rover design.

You should follow this template - it contains descriptions and tips from judges on what to include to get a good score:

Due by 31 May 2026

Must be submitted by 23:59

TEMPLATE for the Project Strategy Report Cars4Mars – ARC 2026

https://docs.google.com/document/d/1VRK_cuNT59AAoXjwiZxBuJfIED1LraKFD2ZqNnilrdM/edit?usp=sharing

You can use both text and graphics (images, diagrams, etc). Format requirements for this document are:

- max 6 pages A4
- font size 11 or more
- margins at least 2.54 cm (1 inch)

Submissions after the deadline, or breaking the format requirements, will result in point penalties, or even zero points being awarded for the report. This is necessary to ensure a level playing field for all the teams.

SUBMISSION LINK for the Project Strategy Report Cars4Mars – ARC 2026:

<https://forms.gle/mzw4d4XV7wZovgRy6>

5.2. Final Design Report

TEMPLATE for the Final Design Report Cars4Mars – ARC 2026

Follow this template - it includes an overview of all the required sections and some tips from the judges on how to get a good score. Make sure to provide all the required sections, especially the "Mars Readiness Review":

https://docs.google.com/document/d/12LH4fIBimjViNcH-vgJYgsDZJJcxG_Hmlv0hG5bBEXs/edit?usp=sharing

Please note: the SUBMISSION LINK WILL BE PROVIDED CLOSER TO THE TIME

Due by 2 August 2026

Must be submitted by 23:59

Format requirements for this document are:

- PDF format
- A4 page size
- max 6 pages A4 (do not exceed this limit - otherwise a scoring penalty will have to be applied to keep it fair towards the other teams).
- font size 11 or more
- margins at least 2.54 cm (1 inch)

Submissions after the deadline, or breaking the format requirements, will result in point penalties, or even zero points being awarded for the report. This is necessary to ensure a level playing field for all the teams.

5.3. Video

The video submission link will be provided before in a later stage.

Record a **max. 5 min** video presenting your team and your rover.

The goal of the video is to show your rover is ready to do well in the Traversal Mission and Autonomous Mission of the Mars Stage.

For example, you'll be expected to showcase:

- Driving forward and backwards
- Turning to avoid obstacles
- Transporting 1kg object in an onboard container
- Driving on difficult terrain (rocks, hills, sand)
- Controlling the rover wirelessly
- Video from the rover cameras being streamed wirelessly to operators
- Your system is able to detect and classify the following objects on the rover cameras: hammer, tennis ball, orange traffic cone.
- Your system is able to detect balloons of different colors on the rover cameras (and their exact position in the frame).
- Your rover drives from one balloon to another autonomously on a flat terrain.

For details, check out the [Mars Stage competition missions](#).

Most points will be awarded to teams that will show the best readiness and good design of their rover. For example, a rover that is driving and also streaming video, will be scored higher than a rover that is only able to drive, but can not stream video. Showing your rover

complete a difficult obstacle course or drive on more difficult terrain will also allow you to score extra points.

Do not get discouraged if your rover is not ready for all the tasks (for example if your rover is not able to drive autonomously). Show as much as you have. Doing well in all aspects of the competition would be really difficult, and ultimately you're only competing against other teams, which will face the same challenges. We expect each rover to be stronger and weaker in different aspects. It is up to you to choose what you focus on. That being said, we really recommend making sure that you can always reliably communicate with the rover and receive live video from it - that was key for the teams that did well in the past edition.

When presenting an action such as driving around obstacles, try to use single-take shots (without cuts), to prove that your rover can work without interruption and make sure that the entire rover is visible. If you are showing that you can control the rover remotely or stream video from the cameras, make sure that we can see you operating the controls and the rover moving at the same time.

Publish the video to YouTube. In the Form for submitting the Final Report there will be a field for pasting the link to the Video.

5.4. Launch Stage Final

This will be in the form of a live presentation that will be held online in front of judges and other teams. Exact requirements will be published at a later stage.

Generally speaking, you will be expected to present:

- Project management methodology
- Your final design
- Lessons you learned during the challenge
- Ideas for improving or commercializing your project

More details will be published at a later stage in the competition.

6. Mars Stage

6.1. Basic info

6.1.1. General requirements

Maximum rover dimensions are:

- 80 cm width
- 80 cm length
- 250 cm height (including any communication masts)

Maximum weight of the rover is 40 kg. These limits do not apply for any equipment inside the command station used for controlling the rover.

Rover needs to be a standalone platform, meaning it can not be attached to anything via cables. It needs to be controlled remotely. You need to have a way to communicate between the rover and your command station (e.g. laptop). We can not guarantee there will be any Wi-Fi on the obstacle course. Make sure to bring your own router if you plan to communicate via Wi-Fi.

Maximum distance that the rover needs to traverse from the command station will not be greater than 80 m.

The rover needs to have its own power source (electrical batteries). Combustion engines and radioactive power sources are **not** allowed in the competition.

Competition missions will not be longer than 30 min, and you can replace the batteries between the missions.

The rover should have an onboard container, to be able to transport objects. These objects will be placed in the container by a judge, and they will always be something that a human can easily pick up with one hand (so not too heavy, not too big).

The rover does not need to have a robotic arm - there is no use for that in the competition missions.

6.1.2. Command station

The rover must be controlled remotely by a human operator, or multiple operators, from your team. The team will get a dedicated desk at the side of the terrain of traversal. The team will have to operate their robot from that desk and can place any equipment they need (laptops, communication antennas). The desk will be equipped with a standard **type M** power outlet (make sure to bring adapters if you need).



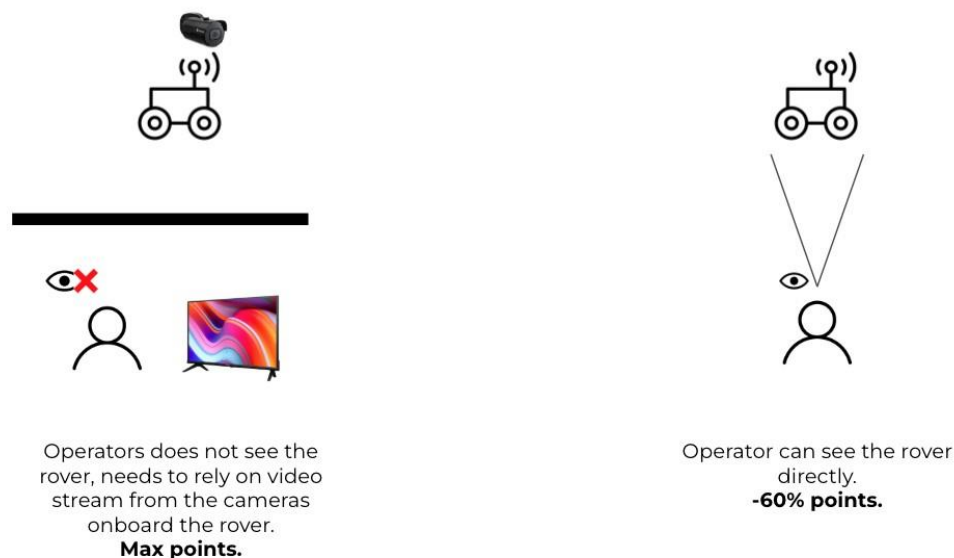
Your team will control the rover from inside a tent placed next to the competition field - like in the photo above.

There are two options for getting visual feedback on the rover - via cameras or directly with the operator's eyes. Each team can choose the option they prefer.

If the team chooses to operate with cameras, then their eye of sight on the terrain will be blocked with a curtain and they will not be able to see their rover with their own eyes (will have to rely purely on cameras). Teams using this method can score maximum points in the missions.

If the team chooses to operate without cameras, they will be able to see their rover directly from the command station or by walking around the obstacle course. **To keep the competition fair, the teams who choose this option will get at least -60% penalty on all the points they score in that particular mission** (e.g. if they score 100 points, they will get 40 points, if they score 50 points, they will get 20 points). We really recommend you focus on getting your cameras to work.

A team can switch from operating with cameras to operating with direct line-of-sight at any point in their mission (but never the other way around). In such a case, the team keeps all the points scored until that moment, and the penalty for direct line-of-sight only applies to all points scored after that.



Being able to control the rover by cameras is the key to doing well in the challenge.

Before the start of the mission, your team will get 10 min to set up all necessary equipment inside the command station (like laptops, monitors, controllers etc.). Radio masts or other communications equipment can be set up max. 10 m from the command station.

After the mission begins, no one is allowed to enter the command station. A team member who would leave the command station can not get back inside it until the end of the current mission.

Any human communication between the operators and people outside the command station will be considered cheating and result in 0 points scored for that task. People inside the command station can only communicate with each other or with a judge who will be present inside the command station through the entire time of the mission.

6.1.3. Interventions

During the competition mission no human can touch the rover (apart from safety interventions). If the rover breaks, the team can call an intervention by communicating this to a judge in the command station. During an intervention team members can touch and repair the rover using any equipment they bring with them. Mission time does not pause during intervention.

6.2. Competition missions

The Mars Stage features 2 rover missions: Traversal Mission and Autonomous Mission. In the time between competition missions, teams can have access to the rovers to repair them, replace batteries, or modify rovers as they see fit. Rovers configurations can be changed between the missions (but each configuration needs to be compliant with the competition requirements).

Important:

The missions are designed to set the bar high and completing everything might be really difficult. This is because each rover will have its own unique strengths and weaknesses, depending on its design. Pick your battles - try to build a rover that can do as much as possible, but remember that you only compete against other teams. The team that completes most wins, which does not mean they need to complete everything - just more than the other teams.

6.2.1 Traversal Mission

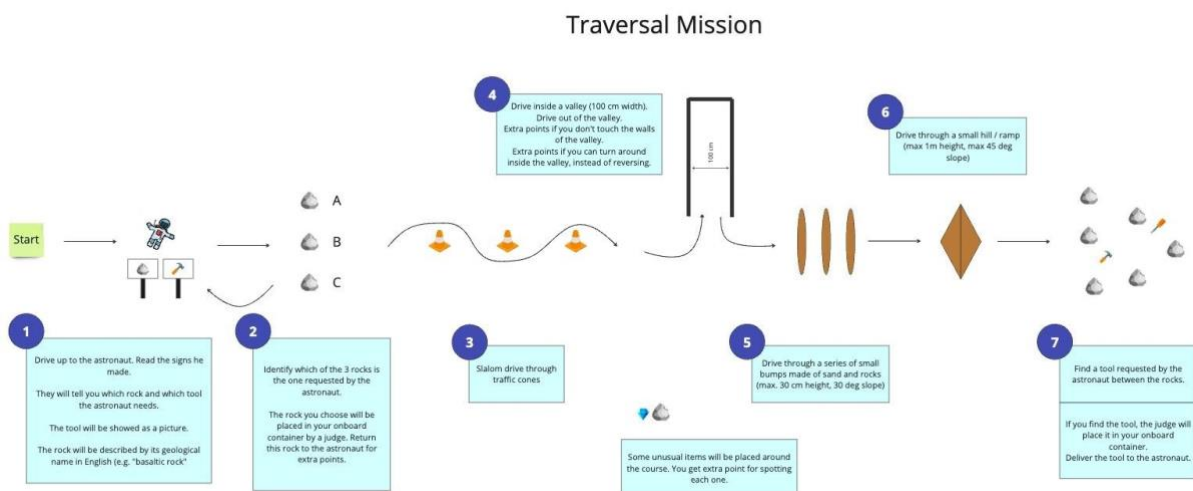


Diagram of the Traversal Mission. For full resolution [click here](#).

During the Traversal Mission, your goal will be to complete a series of tasks - driving through obstacles, looking for objects, transporting objects in your onboard container (placed there by a judge - robotic arm is not needed).

You get points for completing each task. You can skip any task you want and proceed to the next one. In such a case, you don't get points for the task you skipped, and you can not go back to it.

The entire mission is to be completed with the rover that is manually driven by human operators in the command station. There are no extra points for doing anything autonomously.

6.2.2. Autonomous Mission

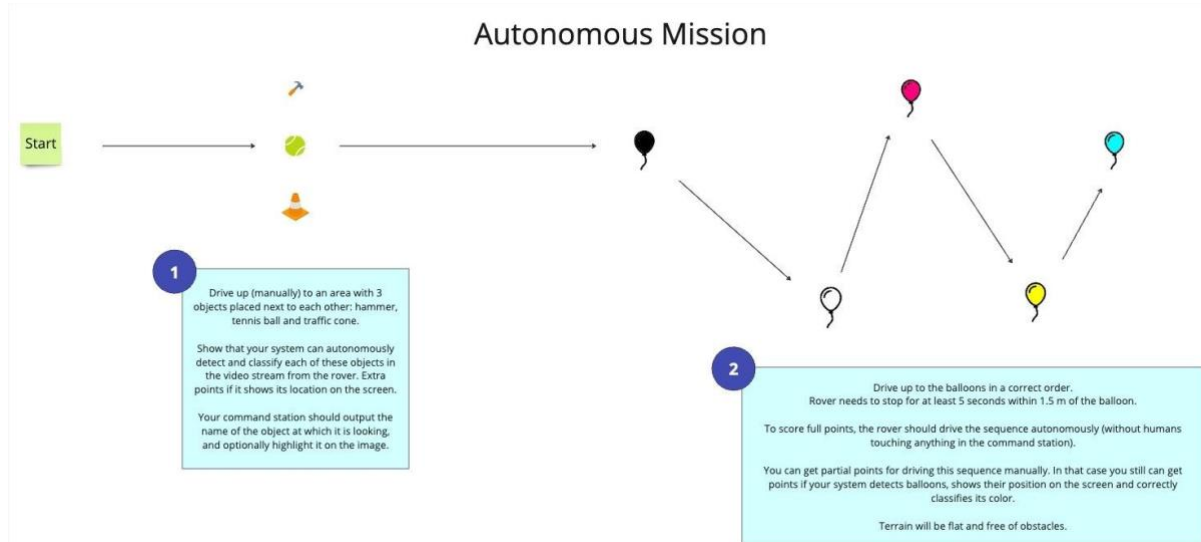


Diagram of the Autonomous Mission. For full resolution [click here](#).

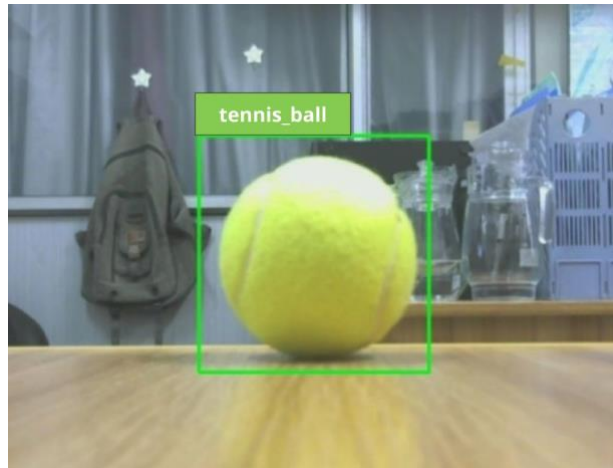
Autonomous Mission allows you to showcase some AI & autonomous capabilities of your rover. Remember, even if your rover is not able to drive autonomously, you can get some points here.

First task is to drive your rover (manually) to a designated place. 3 objects will be placed there next to each other:

- a hammer
- a tennis ball
- an orange traffic cone

You get points if you can show that your system can detect each of these objects in the video stream from your rover, and classify them correctly. You get extra points if your system can show where the object is on the image. The detection system needs to be automatic (AI / computer vision algorithm). A judge will be next to you in the command station to assess that. E.g. your command station can highlight the object in the image and output a message saying "tennis ball".

The computation for detecting the object can be done onboard your rover or in the command station (e.g. laptop) based on the video from the rover - the choice is yours.



Example output of your command station that would count as detection and classification of a tennis ball.

In the next part of the autonomous mission, there will be balloons scattered around the field, attached just above the ground. Colors of the balloons will be: black, white, pink, yellow, blue. These will be standard balloons, such as the ones you would find at a birthday party.

Your task is to drive up to each balloon in correct order:
black -> white -> pink -> yellow -> blue

For each balloon, the rover needs to stop for at least 5 seconds, no further than 1.5 m from the balloon.

To score max points, your rover needs to drive the full sequence autonomously (without operators touching anything in the command station). You get points for each balloon that you reach. The terrain for this task will be flat and without any obstacles, so you do not need obstacle avoidance algorithms.

You can also control the rover manually the whole time. You still get points for reaching each balloon, just much less than if you do it autonomously. If you can at least show that your system correctly detects the location and type of the balloon in the image, you still get some partial points.

7. Tips from the judges

1. Make sure you read and understand the entire Rulebook, to make sure you design your rover for the actual missions and complete all the deliverables along the way. If something is unclear - just write to us, we'll be happy to clarify!
2. Start by assembling a team of people who are passionate to work on the rover and bring the necessary skills to the table. You'll spend quite some time with these folks, so make sure you fit together well, so that you can enjoy the ride!
3. You don't need to know how to build a rover yet - as long as you have the passion to do proper research and learn along the way. This is what this challenge is for.

4. Project management is key. Make sure you clearly split responsibilities between team members and that you communicate often on the progress and challenges. Plan a full timeline from today to the day of the competition with some realistic goals along the way. This way you will know if you are behind schedule. If that happens don't panic - it's perfectly normal in every engineering project. Maybe you'll have to skip some features of the rover to be on time (e.g. skip the AI features if your rover still does not have a proper video stream). Be smart about what you prioritize and what you skip - this is also a key skill for every future engineer.
5. Find people who will support you! Your university or school, sponsors, advisors, government. Your ambition will be inspiring for many, and you might be surprised how many people and companies might be willing to support you (this includes collecting funds, parts and tools for building the rover, and also the funds for travelling to the Mars Stage in Johannesburg).
6. Test often, test early. Don't just try to build a complete rover and run it at the end - this won't work. Make sure you test every new component you add.
7. Focus on the key features of the rover: it's ability to drive, remote control, video stream, batteries.
8. Don't forget you need to control your rover remotely somehow. We provide a power outlet, and that's about it. If you need to connect to your rover via Wi-Fi, bring your own router. If you need internet connection, make sure you have that covered.